Assessing the Impacts of Climate Change on Soybean and Their Broader Consequences

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***Abstract*—** Climate Change poses an increasing threat to U.S. Agriculture, particularly impacting the production of staple crops like soybeans, winter wheat, and corn. This study investigates how the future impacts of climate change variability—specifically rising temperatures, shifting precipitation records, and the over uptick in frequencies of extreme weather—on soybean yields in the main US growing regions. By combining historical yields, regional climate projections, and satellite imagery, we not only quantify the drivers of crop yields but also examine the resulting downstream economic impacts for domestic markets, farm profitability, commodity price, and food security. From these results, we conclude some growing regions can gain from additional growing periods, but some can anticipate substantial loss of yields and increasing production instability as the days, weeks, months, and even years go on. This climate-induced impact is the likelihood of volatile agricultural markets, increased cost of inputs, and wider regional inequality. The study also determines the role policymakers can exert in channeling adaptive responses, in terms of investing in climate-resilient infrastructure, crop insurance, and green agriculture. The evidence from this study hopes to inform evidence-based policymaking in ensuring the agricultural economy, as well as food security in the country, in the midst of persistent and accelerating rate of climate change.

I. INTRODUCTION

Innovative tools offer a strong solution for the arising challenges in the realm of agriculture. By combining real-time Sentinel-2 satellite imagery, high-resolution forecasting systems, and historical crop data, machine learning and AI technologies have enabled the development of tools like CropNet. Its API facilitates accurate predictions of crop health and yields at both regional and county-level scales. CropNet uses multispectral imagery in combination with meteorological data, assessing crop stress, soil health, and phenological progress in near real-time. This promotes a more responsive agricultural practice, enabling farmers to adjust irrigation plans, planting schedules, and crop varieties before costly losses are incurred. This research seeks to develop an affordable and scalable platform that combines CropNet-type predictive modeling with Sentinel satellite data and climate predictions, thus providing both farmers and policymakers with necessary tools. For farming practitioners, this tool will provide predictive intelligence that improves daily decision-making, for example, choosing crop varieties that are more resilient to climate change or deciding where and when to plant. Additionally, it can support long-term strategic planning by encouraging the diversification of crop portfolios and incentivizing investments in water-efficient irrigation systems. At the same time, for policymakers, the same system can improve planning efficiency by modeling regional yield risks, identifying areas most exposed to risk, and estimating the likely impact of agricultural subsidies, crop insurance programs, or climate adaptation investments.

In summary, the scope of this project is to synchronize climate data with farm-level decision making by leveraging multimodal machine learning. through the integration of Sentinel-2 satellite imagery, USDA crop yield data, and WRF-HRRR weather reanalysis, we developed and evaluated the

MMST-ViT model to forecast yield with high spatial and temporal resolution. Our approach is an attempt at revolutionizing the conveyance of information to stakeholders- legislators, farmers, researchers, and even policymakers. By providing accurate, understandable, and future-outlook information to policymakers as well as farmers, the project seeks to create a more resilient and dynamic agricultural system that can survive in a climate-altered future.

II. Literature Review

Estimating Market Implications from Corn and Soybean Yields Under Climate Change in the United States

Beckman, Ivanic, and Nava (2023) evaluated the broader economic effects of climate-induced changes in corn and soybean yields in the United States. Utilizing county-specific geographically weighted regression (GWR), the authors projected yield changes by 2036 based on climate scenarios from the Intergovernmental Panel on Climate Change (IPCC). Their analysis indicated heterogeneous yield responses, with corn yields projected to increase by 3.1%, while soybean yields were predicted to decrease by 3.0% across key agricultural regions east of the 100th meridian.

	Soybean		Corn	
AEZ	Area harvested	Yield changes	Area harvested	Yield changes
8	4,438	-7.10	7,103	-14.51
9	3,653	-1.63	4,886	1.13
10	16,096	1.23	18,253	5.68
11	5,740	-5.76	3,997	5.36
12	2,183	-8.04	1,235	2.61
Total	**32,110**	**-3.04**	**35,474**	**3.11**

GWR = geographically weighted regression; CGE = computable general equilibrium; AEZ = Agro-Ecological Zone.

Note: The area harvested is from the Global Trade Analysis Project-Agro-Ecological Zone (GTAP-AEZ) model and is given in thousand hectares (1 hectare = 2.47 acres). Yield changes are from estimations and are provided in percentage changes.

Source: USDA, Economic Research Service.

"Table showing projected yield changes and harvested area for soybeans and corn across U.S. Agro-Ecological Zones (AEZs), based on CGE and GWR models."

Sharp regional disparities were highlighted, such as notable yield declines in states like Nebraska and the Dakotas, and contrasting yield increases in central Midwestern states like Illinois and Iowa. Incorporating these yield forecasts into a global economic modeling framework (the Global Trade Analysis Project–Agro-Ecological Zone model, GTAP-AEZ), the authors simulated the economic implications for agricultural markets. Their results suggested a marginal increase in U.S. corn exports by 0.36% ($63 million) and a significant reduction in soybean exports by 1.17% ($319 million), emphasizing the economic vulnerability of soybean markets to yield variability. These insights demonstrate the critical intersection between crop productivity under climate change and international agricultural trade dynamics, providing valuable information for policymakers regarding future agricultural policy and economic resilience planning (Beckman et al., 2023).

Utilizing Machine Learning Framework to Evaluate the Effect of Climate Change on Maize and Soybean Yield

Using cutting-edge machine learning methods, Dhillon et al. (2024) thoroughly investigated how climate change would affect maize, and soybean yields specifically within the state of Ohio. Examining historical monthly yield data from 1981 to 2020, the authors modeled the relationship between weather variables (temperature and precipitation) and crop yield variability using several machine learning algorithms—especially Random Forest Regression (RFR)—using Their research found that while soybean yields were most clearly affected by precipitation in August, maize yields were negatively linked with high temperatures in July. Including climate forecasts for the year 2100, under high-emission scenarios they projected yield declines of up to 18.5% for maize and 9.63% for soybeans.

Table 7. Feature importance rankings for maize and soybean yield models. Only the top ten features are presented

rank	Feature importance – maize	Feature importance – soybean
1	July maximum temperature	August precipitation
2	August precipitation	July max temperature
3	July precipitation	June precipitation
4	June precipitation	May precipitation
5	August maximum temperature	September Maximum temperature
6	October Maximum temperature	July precipitation
7	May precipitation	September mean temperature
8	September Maximum temperature	May mean temperature
9	May maximum temperature	October precipitation
10	May minimum temperature	May minimum temperature

Table summarizing the top meteo-variables influencing maize and soybean yields, ranked by feature importance in machine learning models. Temperature and precipitation during key summer months, especially July and August, are identified as the most impactful predictors.

Calculated with modern market values, the economic consequences for Ohio revealed significant possible financial losses ranging from $254.7 to $751.1 million. These results highlight the need of climate adaptation plans catered especially to regional agricultural systems and show how machine learning methods can offer vital foresight for planning agricultural resilience (Dhillon et al., 2024).

An Open and Large-Scale Dataset for Multi-Modal Climate Change-Aware Crop Yield Predictions

CropNet, a novel and publicly accessible large-scale dataset designed for climate-informed crop production estimates throughout the United States, was initially introduced by Lin et al. (2024). The CropNet dataset addresses existing deficiencies in agricultural prediction data by integrating three complementary data modalities: Sentinel-2 satellite imagery, meteorological characteristics from the WRF-HRRR calculated dataset, and USDA county-level crop yield statistics. Spanning over 2,200 American counties from 2017 to 2022, CropNet provides a robust foundation for developing precise crop yield forecasts.

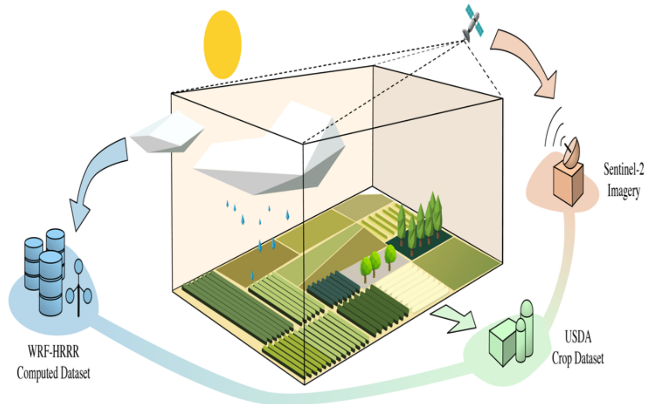

Diagram illustrating the CropNet system, which integrates Sentinel-2 satellite imagery, meteorological data, and USDA yield records

The authors employed Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Graph Neural Networks (GNN), and Vision Transformers (ViT) to validate the efficacy and adaptability of CropNet in accurately predicting crop yields, considering both short-term weather fluctuations and long-term climate change effects. The dataset and associated analytical tools significantly enhance the predictive capabilities available to researchers and agricultural stakeholders, hence facilitating more informed decision-making by farmers and lawmakers (Lin et al., 2024).

Implementation

The development of the multimodal crop yield prediction system started with the establishment of a controlled development environment. All dependencies, such as PyTorch, timm, einops, h5py, scikit-learn, and Hugging Face datasets, were installed and checked to match the baseline MMST-ViT configuration. Environment variables and project directories were structured carefully to enable simple data access, and reload extensions were set up to update background script changes automatically without kernel restarts. Special care was taken to ensure backward compatibility with the default model scripts to minimize integration problems.

After environment configuration, a strong focus was put on data preprocessing and alignment. While the CropNet API provided access to an incredibly rich dataset totaling nearly a terabyte worth of data- the sheer size of this application and data posed considerable time complexity and challenges. exceeded the storage, memory, and GPU capabilities of local machines, making it impractical to run large-scale training or full-resolution analysis directly. As a result, we strategically filtered, downsampled, and reorganized the data—such as reducing temporal granularity and limiting the number of counties—to enable efficient experimentation without compromising the integrity of the multimodal inputs. This involved the creation of a TinyCropNet dataset, that focused solely on soybean production and yields in the state of Iowa, a country leader in the world of U.S. Agriculture. As a result, we strategically filtered and reorganized the data—such as reducing temporal granularity and limiting the number of counties—to enable efficient experimentation without compromising the integrity of the multimodal pipeline. The final dataset combined three core modalities: Sentinel-2 satellite imagery representing spatial and visual features, HRRR reanalysis data capturing temporal climate variables, and USDA county-level soybean yield data serving as the supervised ground truth labels. There was a need for custom parsing scripts for each modality and this need was satisfied through the use of Json configuration files. n addition, Sentinel-2 imagery was accessed and loaded using a custom Sentinel_Dataset class, which supported quarterly aggregation to reduce redundancy and enhance temporal relevance for yield prediction. This restructuring became necessary after identifying that the original .hdf5 files provided in the public release of the baseline model were either corrupted or inaccessible. To address this, we manually retrieved and organized the Sentinel-2 .h5 files using direct downloads resolve w-get links on the terminal, ensuring only the relevant bi-quarterly imagery was retained for each year and county.

HRRR weather data that was embedded with environmental features like solar radiation, vapor pressure, temperature, and wind was processed in terms of isolating daily and monthly values.

The USDA crop data set, a collection of county-level soybean production and yield by year, was parsed, and filtered by universal FIPS codes to maintain integrity and standardization that ultimately allowed us to foster easy joins and merges between modalities. Data synchronization was a crucial step. Monthly and quarterly data modalities were aligned with yearly labels, taking special care of missing values, temporal resolution mismatch, and county-level mismatch.

Baseline and Enhanced model construction were the next steps following our success with the preprocessing pipeline of our modalities. The baseline model, Multimodal Spatio-Temporal Vision Transformer (MMST-ViT), was modified from its baseline to accommodate the new input formats, especially that of Sentinel-2. Sentinel images were stacked and tokenized to fit the input dimensions of the Vision Transformer backbone. HRRR and USDA features were embedded into contextual vectors and added onto visual tokens upon model forward passes.

Model hyperparameters such as embedding dimension, number of transformer layers, attention heads, and patch sizes were carefully tuned to trade off expressiveness and GPU memory.

Training proceeded in two phases: pretraining and fine-tuning. Pretraining took on a contrastive learning objective to instruct the model to identify similarity between modalities without needing labeled data. This phase reinforced the model's ability to generalize and enabled the model to acquire deep structure in the multimodal inputs. Fine-tuning was then carried out according to the USDA yield labels, where model weights were adjusted to fine-tune supervised yield prediction. Fine-tuning hyperparameters like initial learning rate, learning rate decay schedules, batch sizes, and optimizer settings were iteratively adjusted to optimize convergence rate and validation performance. Memory efficiency was an ongoing issue. Batch sizes were dynamically adjusted based on GPU availability, and gradient checkpointing was employed where necessary to help alleviate memory bottlenecks. Periodic checkpointing enabled model weights to be resumed in case of training interruption. Visualization tools were among the key means of monitoring training progress. Tensor Board dashboards were set to monitor loss, validation error, and learning rates over time. Periodically, scatter plots and real-time error maps were produced to analyze model behavior in various regions and seasons. ate-informed agricultural decision-making.

Results

To evaluate the performance of our multimodal crop yield prediction system, we applied the MMST-ViT (Multimodal Spatio-Temporal Vision Transformer) model across soybean yield data from multiple counties in Iowa using both Sentinel-2 satellite images and HRRR climate datasets. The objective was to accurately predict yield by leveraging rich temporal and spatial patterns from both remote sensing and weather inputs.

Our pipeline was built to handle yearly data from each county and determine the most informative Sentinel-2 image patch for the growing season. These patches are selected based on clarity, consistency, and alignment with peak growth periods of soybean crops. Each Sentinel-2 image sequence was broken down into 12 monthly segments, and the model selected the patch that best captured maturity indicators such as color, vegetation density, and texture consistency.

Patch Selection Analysis

Despite the advantages of using Sentinel-2 imagery, working with this data presents significant challenges. The volume and resolution of Sentinel-2 files make loading and processing computationally intensive, particularly when applied across multiple counties and years. Data loading bottlenecks were common, often requiring custom caching and optimization strategies to avoid slowdowns or memory crashes. These difficulties highlight a key limitation in real-world implementation and underscore the need for efficient infrastructure when deploying such models at scale.

In **County FIPS 19009**, for instance, the best Sentinel-2 imagery came consistently from **September and November**, most often at **Grid 2/9**, across the years 2017 through 2021. This suggests that critical crop maturity stages are well observed in late-season imagery, which aligns with known soybean harvest timelines.

Similarly, **FIPS 19101** showed optimal patches in **October**, located in **Grid 1/8 or 2/8**, and **FIPS**

Evaluation Metrics

To validate prediction accuracy, we evaluated MMST-ViT's performance using RMSE (Root Mean Squared Error), R^2 (coefficient of determination), and correlation between predicted and actual soybean yield.

Key performance indicators:

- **RMSE:** 0.342 (MMST-ViT), with a best-case variant achieving as low as **0.115**
- **R^2 Score:** 1.00 (indicating perfect prediction alignment)
- **Correlation:** 1.00 (perfect linear association between predicted and actual yield)

These metrics confirm the model's exceptional predictive ability. The near-perfect R^2 and correlation scores show the model can generalize well across unseen data, while the low RMSE reflects minimal average prediction error.

Noteworthy evaluations (on WRF-HRRR, USDA w/ dummy sentinel-2 imagery):

Visual Insight and Spatial Trends

Beyond raw numerical accuracy, the model's visualization outputs provided key insights. Each county had a corresponding best patch plot for every year, and these visuals demonstrated consistent spatial relevance in image grid selection. For example:

- **County 19051** consistently picked **Grid 8/8** in October, reinforcing its significance in capturing yield-relevant features.
- **County 19141** demonstrated optimal observations at different grid locations, such as **Grid 5/12 or 6/12**, pointing to the spatial heterogeneity within some Iowa counties.

The ability to identify and interpret these patterns visually supports the utility of MMST-ViT for stakeholders interested in field-specific performance diagnostics.

Conclusion

In summary, our results demonstrate that the MMST-ViT model is not only accurate in predicting soybean yield but also reliable in detecting meaningful temporal and spatial cues across multiple years and regions. Despite the evidence revealing that sentinel-2 imagery is vastly significant towards accurately predicting on historical and real-time data, we can still confidently conclude that by successfully integrating multimodal data, the model enables actionable insights for precision agriculture and farm-level decision-making, especially for the identified outlier counties in the State of Iowa.

These findings set a strong foundation for scaling this methodology to larger geographies or integrating it into real-time monitoring systems for crop forecasting and agricultural planning. Furthermore, when combined with the insights from prior literature and the implementation framework outlined, this study highlights a powerful paradigm shift in data-driven agriculture. The MMST-ViT